Polymorphous Light Eruption (PMLE)

Clinical Features

History

Polymorphous light eruption (PMLE) typically presents in the first three decades of life and affects females two to three times more frequently than males. It occurs across all skin types and racial groups but is more prevalent in Caucasians. Approximately 20% of patients report a positive family history, suggesting a genetic predisposition.

The eruption usually appears each spring or during sunny vacations, following the first significant ultraviolet radiation (UVR) exposure after a period of minimal sun exposure. It may also be triggered by recreational sunbed use or, very rarely, by visible light. The condition often diminishes with continued sun exposure throughout the season—a phenomenon known as "hardening" or desensitization. PMLE is uncommon in winter unless exposure occurs via UVR reflected from snow or transmitted through window glass.

Individual susceptibility varies. The time between UVR exposure and onset of lesions ranges from hours to days, typically not less than 30 minutes (though itching may precede visible changes). Lesions resolve completely within days to one or two weeks after exposure ends, without scarring. Recurrent outbreaks in the same individual tend to affect the same sun-exposed areas symmetrically, although only some exposed skin is usually involved—particularly areas that are typically covered during winter. Large portions of exposed skin may remain unaffected.

An extreme variant, **juvenile spring eruption**, primarily affects boys in spring and is characterized by pruritic papules and vesicles localized to the ear helices; typical PMLE may coexist. Systemic symptoms such as malaise or nausea are rare but possible in isolated cases.

Cutaneous Lesions

PMLE exhibits a wide range of morphologic presentations, all sharing similar pathogenesis and prognosis. Common forms include: - **Papular** - **Papulovesicular** - **Plaque-like**

Lesions are pruritic, erythematous, and symmetrically distributed, primarily affecting sun-exposed sites. The rash develops within hours of UVR exposure and resolves fully over several days to two weeks.

Pathogenesis

PMLE is very likely a genetically determined **delayed-type hypersensitivity (DTH) reaction** directed against UVR-induced cutaneous antigens. Histologically, it shows **epidermal spongiosis** with a **dermal perivascular infiltrate** composed predominantly of mononuclear cells and associated edema.

The specific radiation-absorbing molecules that initiate PMLE have not been identified, but multiple candidates may be involved, varying between and within individuals. Potential mechanisms include: - Direct antigenicity of UVR-altered molecules - Free radical-mediated modification of nearby cellular components to form neoantigens
Both mechanisms may contribute.

Repeated sunlight exposure induces cutaneous immunologic tolerance, which may suppress the underlying DTH response—explaining the progressive desensitization ("hardening") observed as summer progresses or during prophylactic phototherapy.

Ultraviolet Radiation Sensitivity

UVA radiation (315–400 nm) is generally more effective than UVB (280–315 nm) in triggering PMLE. Studies show: - UVA elicits the eruption in approximately **56%** of patients - UVB in about **17%** - Combined UVA/UVB in **27%**

Other reports suggest UVB sensitivity may occur in up to **57%** of patients. Overall: - ~**50%** of PMLE patients are sensitive to UVB - ~**75%** are sensitive to UVA - ~**25%** react to both

Visible light may rarely play a role.

Notably, many sunscreens block UVB effectively but transmit UVA, potentially leading to a **paradoxical worsening** of PMLE despite sunscreen use.

Management

PMLE responds to: - **Broad-spectrum sunscreens** (with strong UVA protection) - **Short-term topical or oral corticosteroids** - **Prophylactic low-dose immunosuppressive phototherapy** (e.g., narrowband UVB or UVA) initiated in spring for individuals with frequent recurrences

This prophylactic approach leverages the natural development of tolerance, reducing symptom severity over time.